Project-Presentation

Course: [ME444] – Analysis and Design of Mechanical Systems

Department of Mechanical Engineering

Data-Driven Active Vibration Control using Neural MPC in a ROS-Gazebo Simulation Framework

Presented By

24B2146 Sahil Kadale
24B2258 Aniket Pol
24B2259 Vatsh Van
24B2260 Anmol Raj

Under the supervision

Prof. P. Seshu (Department of Mechanical Engineering)

INTRODUCTION

Problem Statement:

Mechanical systems subjected to base excitation experience vibrations that degrade performance and stability. Existing approaches primarily focus on passive isolation and offline frequency-domain analysis, lacking real-time adaptability.

Implementation Idea:

- Develop a simulation-based active vibration control system
- Implement Kalman filtering for accurate state estimation
- Design a learning-based optimization model (Neural MPC approximation)
- Deploy a closed-loop control system in ROS-Gazebo with multi-actuator setup to minimize vibration amplitude under dynamic excitation

Reference: Varanis et al. (2016); Neural MPC (Salzmann et al.); Cao et al. (2020)

METHODOLOGY

- **Signal Generation:**
  Simulated vibration signals (acceleration, velocity, displacement) with sampling consistent with vibration analysis standards (~200–250 Hz).

- **State Estimation:**
  Kalman Filter used to estimate system states from noisy measurements.

- **Optimization Model:**
  Learning-based control policy: $u = \pi(x)$, where x = estimated state, u = actuator input
  Objective: $\min \Sigma\,(\text{vibration})^2 + \lambda\,|u|^2$

- **Simulation Environment:**
  ROS + Gazebo platform with 4 actuators for active vibration control.

- **Closed-loop Control Pipeline:**
  Sensor → Kalman Filter → Neural Model → Actuator Commands → Simulation Feedback

Reference: Varanis et al. (2016); Neural MPC (Salzmann et al.); Cao et al. (2020)

WORK PLAN

Mechatronics:

- Generate vibration signals
- Implement Kalman filtering
- Perform time-domain and frequency-domain (FFT) analysis
- Record training dataset (state-action pairs)

Optimization Model:

- Build the optimization model based on physics of spring-mass damper system
- Train neural network for control policy for accounting physics approximations and to give warm start to MPC
- Approximate MPC-based control

ROS + Simulation:

- Develop Gazebo environment with 4 actuators
- Implement ROS nodes and communication pipeline Integrate simulation physics

Control Architecture & Integration:

- Define system-level control loop
- Integrate all modules (sensor → model → control)
- Evaluate system performance and stability

Reference: Varanis et al. (2016); Neural MPC (Salzmann et al.); Cao et al. (2020)

PERMISSIONS AND RESOURCES

Resources Required:

- *Software:* Python (NumPy, SciPy, PyTorch)
  ROS (Robot Operating System)
  Gazebo Simulation
- *Computational Resources:* Standard computing systems for simulation and training
- *Hardware:* GY-521 board with accelerometer MPU-6050
  Arduino Mega 2560 Rev3 Board
  HC-SR04 distance sensor

Permissions:
No laboratory or fabrication permissions required.

Justification:
Simulation-based approach eliminates high-cost experimental setups while enabling scalable and controlled testing, as experimental vibration systems require complex instrumentation.

Reference: Varanis et al. (2016); Neural MPC (Salzmann et al.); Cao et al. (2020)

EXPECTED OUTCOMES AND TIMELINES

Expected Results

- **Dynamic Analysis:**
  - Position, velocity, acceleration signals
  - Frequency-domain analysis using FFT
- **Control Performance:**
  - Reduction in vibration amplitude
  - Improved system stability under disturbance
- **Simulation Validation:**
  - Comparison of controlled vs. uncontrolled system
  - Multi-actuator coordination performance

Timeline of Work

- **Week 1–2:**
  - Signal modeling
  - Kalman filter implementation
  - Data Collection
- **Week 3–4:**
  - Optimization model training
  - ROS-Gazebo simulation setup
- **Week 5:**
  - Integration and closed-loop testing
  - Performance evaluation and report

THANK YOU!!!